

# Introduction to mathematics for economists

## Exercise Sheet #3

Alexandre Grellet

2026-08-28

During the tutorials, we will solve indefinite integrals 8 and 11, definite integrals 1 and 3, the second-price auction exercise, the waiting-time exercise, and the wage-bargaining exercise. Prioritise these. The **discrete lottery** exercise is a recommended warm-up before the continuous probability exercises. A full answer sheet is provided, since we will not have time to solve everything in class.

## Indefinite Integrals

### Indefinite Integral

By **indefinite integral**, we mean an antiderivative (or primitive) of the function  $f$ . For instance, the antiderivative of  $f(x) = x$  is:

$$\int x \, dx = \frac{x^2}{2} + C$$

where  $C$  is just a constant. We call it *indefinite* because we do not specify the bounds — we only look for the primitive.

Compute the following indefinite integrals (do not forget to specify the definition set). Hints are given for the ones that need a technique beyond the standard table of antiderivatives.

1.  $\int x^\alpha \, dx$ , with  $\alpha \in \mathbb{R}$ .
2.  $\int x^2 e^{-x} \, dx$ .
3.  $\int x \ln(x) \, dx$ .
4.  $\int \cos^2(x) \, dx$ .
5.  $\int \frac{x}{x^2 + 1} \, dx$ .
6.  $\int \frac{1}{x^2 - 1} \, dx$ . *Hint: you may use  $\frac{1}{x^2 - 1} = \frac{1/2}{x - 1} - \frac{1/2}{x + 1}$*
7.  $\int (ax + b)^n \, dx$ , with  $a \neq 0$ ,  $b \in \mathbb{R}$ ,  $n \in \mathbb{N}$ .
8.  $\int \frac{\ln(x)}{x} \, dx$ .
9.  $\int \frac{\ln^n(x)}{x} \, dx$ , with  $n \in \mathbb{N}$ .

10.  $\int \frac{2x+1}{x^2+x+1} dx.$

11.  $\int \frac{1}{1+x^2} dx.$  (*Hint: recall that  $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}.$* )

## Definite and Improper Integrals

Compute the following integrals, when they are well defined.

1.  $\int_1^{\infty} \frac{1}{x^2} dx.$
2.  $\int_1^{\infty} \frac{1}{x} dx.$
3.  $\int_0^{\infty} e^{-x} dx.$
4.  $\int_e^{\infty} \frac{1}{x(\ln x)^2} dx. \text{ (Hint: substitution } u = \ln x.)$
5.  $\int_{-1}^1 \frac{1}{x} dx$

## Leibniz Rule and Applications

Let

$$F(t) = \int_t^{t^2} \frac{x^2}{2} t \, dx.$$

Here the integrand  $f(t, x) = \frac{tx^2}{2}$  depends on  $t$ , **and** both bounds  $a(t) = t$ ,  $b(t) = t^2$  depend on  $t$ . Assume  $t > 1$ .

1. Compute  $F'(t)$  **without** computing the integral first.
2. Compute the explicit value of the integral, then differentiate it.
3. Check that the two answers agree.

## Second-Price Auction

In this exercise, you will see how the Leibniz rule can be used in an economic context: the **second-price (Vickrey) auction**, where the highest bidder wins but pays the *second-highest* bid. We will show that bidding your true value is optimal and, remarkably, this holds *whatever* the other bidder does.

Bidder  $i$  has private value  $v_i$  (i.e. individual  $i$  is ready to pay up to  $v_i$  to acquire the good) and submits a bid  $b_i \geq 0$ . The opponent submits a bid  $B$ , which from  $i$ 's point of view is a random variable with density  $g$  and CDF  $G$  on  $[0, \bar{b}]$  (think of  $\bar{b}$  as the upper possible value for the good).

1. Explain why bidder  $i$ 's expected payoff from bidding  $b_i$  is

$$u_i(b_i) = \int_0^{b_i} (v_i - x) g(x) \, dx.$$

2. Differentiate  $u_i$  with respect to  $b_i$  (use Leibniz rule for a moving upper bound), and deduce the optimal bid.

## Probability: Expectations and Variance

### Discrete Random Variables

A **discrete random variable**  $X$  takes countably many values  $x_1, x_2, \dots$  with **probability mass function**  $p(x_i) = \Pr(X = x_i) \geq 0$  and  $\sum_i p(x_i) = 1$ . Its **expectation** and **variance** are

$$\mathbb{E}[X] = \sum_i x_i p(x_i), \quad \text{Var}(X) = \sum_i (x_i - \mathbb{E}[X])^2 p(x_i) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

For any constants  $a, b$ :  $\mathbb{E}[aX + b] = a \mathbb{E}[X] + b$  and  $\text{Var}(aX + b) = a^2 \text{Var}(X)$ .

### A Discrete Lottery (warm-up)

You invest \$10 in a start-up. Your net payoff  $X$  (in dollars) is:

$$\Pr(X = -10) = 0.6, \quad \Pr(X = 10) = 0.3, \quad \Pr(X = 60) = 0.1,$$

i.e. the venture usually fails (you lose your \$10), sometimes returns \$10, and rarely delivers a big \$60 success.

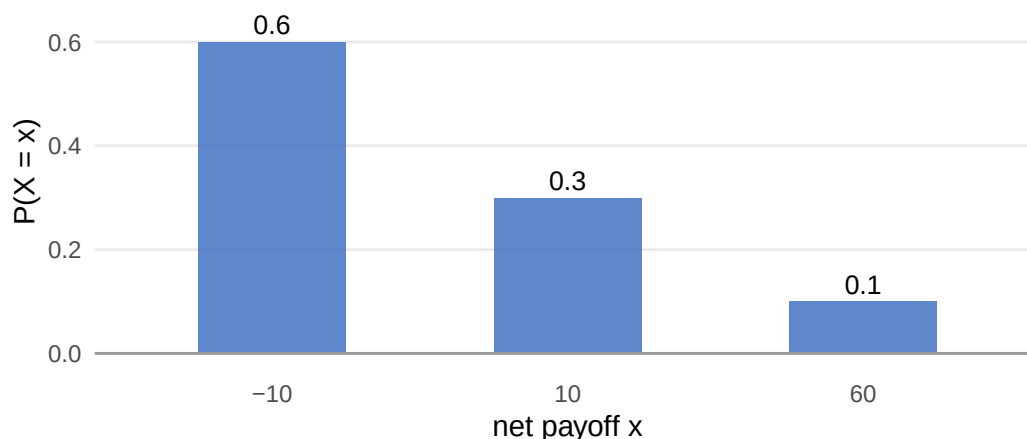


Figure 1

1. Check that this is a valid probability mass function.
2. Compute the expected net payoff  $\mathbb{E}[X]$ .
3. Compute  $\text{Var}(X)$  and the standard deviation  $\sigma$ .
4. A fund lets you take **twice** (\$20 in the startup) the position for a flat \$2 fee, so your payoff becomes  $Y = 2X - 20$ . Use the linearity rules to find  $\mathbb{E}[Y]$ ,  $\text{Var}(Y)$  and  $\sigma_Y$ .
5. A safe government bond returns \$13 for sure. Compare its mean and variance with those of  $X$ . Which would a **risk-averse** investor prefer, and why?

## Continuous Random Variables

A **continuous random variable**  $X$  is described by a **probability density function** (PDF)  $f : \mathbb{R} \rightarrow \mathbb{R}_+$  satisfying:

$$\int_{-\infty}^{+\infty} f(x) dx = 1.$$

The probability that  $X$  falls in an interval  $[a, b]$  is:

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx.$$

The **cumulative distribution function** (CDF) is defined as:

$$F(t) = \Pr(X \leq t) = \int_{-\infty}^t f(x) dx.$$

The **expectation** (mean) of  $X$  is:

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx.$$

The **variance** of  $X$  measures the spread around the mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int_{-\infty}^{+\infty} (x - \mathbb{E}[X])^2 f(x) dx = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

## Waiting Times at a Service Desk

A firm operates a customer service desk. The waiting time  $T$  (in minutes) before a customer is served is modelled as a continuous random variable with probability density function:

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0,$$

where  $\lambda > 0$  is a parameter capturing how fast the desk operates (a higher  $\lambda$  means shorter average wait). This is the **exponential distribution** with parameter  $\lambda$ .

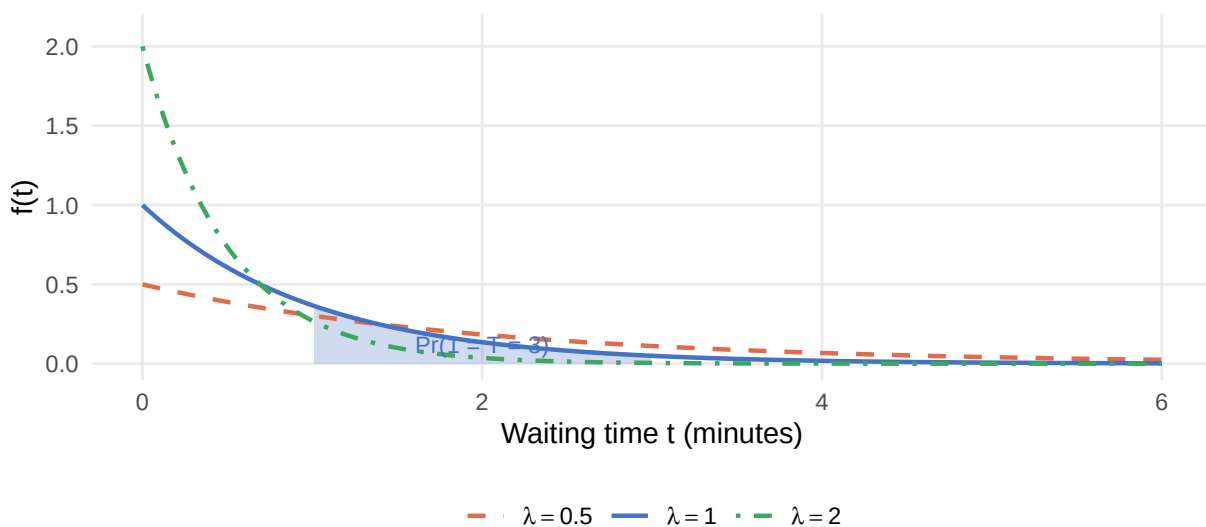


Figure 2

1. Verify that  $f$  is a valid PDF, i.e. show that  $\int_0^{+\infty} \lambda e^{-\lambda t} dt = 1$  and it's never negative.
2. Compute the CDF  $F(t) = \Pr(T \leq t)$ .
3. A customer is “well served” if they wait less than 3 minutes. A manager sets  $\lambda = 1$ . What is the probability that a customer waits more than 3 minutes?
4. Compute  $\mathbb{E}[T]$  and interpret it in terms of the firm's service desk.
5. Compute  $\text{Var}(T)$  using  $\text{Var}(T) = \mathbb{E}[T^2] - (\mathbb{E}[T])^2$ . (*Hint: use integration by parts*)

## General economic applications

### Consumer and Producer Surplus

#### Consumer and Producer Surplus

Consider a market with inverse demand  $P_D(q)$  and inverse supply  $P_S(q)$ , and let  $q^*$  be the equilibrium quantity at price  $p^*$ .

- The **consumer surplus** is the aggregate gain to buyers:

$$CS = \int_0^{q^*} (P_D(q) - p^*) dq.$$

- The **producer surplus** is the aggregate gain to sellers:

$$PS = \int_0^{q^*} (p^* - P_S(q)) dq.$$

Consider a market for a consumption good with the following inverse demand and supply curves:

$$P_D(q) = 12 - q^2, \quad P_S(q) = 2q + 2.$$

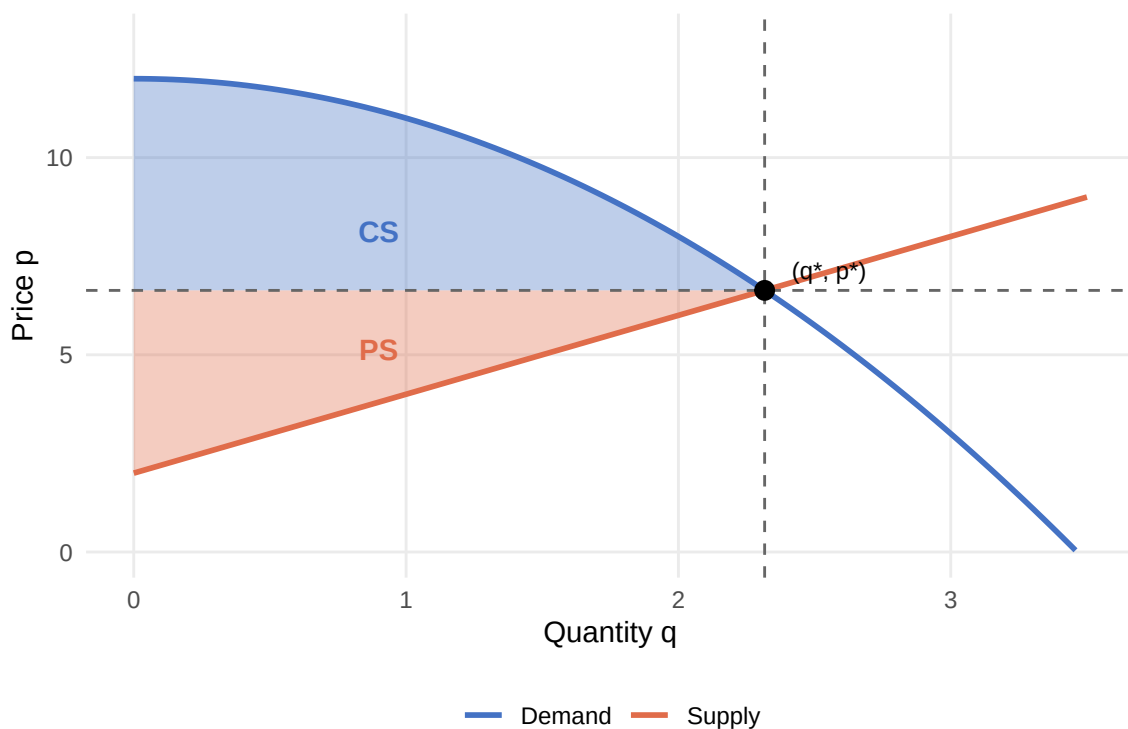


Figure 3

1. Find the equilibrium price  $p^*$  and quantity  $q^*$  (*hint: supply must equal demand*).
2. Compute the consumer surplus  $CS$ .
3. Compute the producer surplus  $PS$ .
4. What is the total surplus  $TS = CS + PS$ ? Briefly interpret this number.



## Lorenz Curve and Inequality

### Lorenz Curve and Gini Coefficient

Let  $F$  be the CDF of income in a population, with mean  $\mu$ . The **Lorenz curve**  $L : [0, 1] \rightarrow [0, 1]$  maps the cumulative population share  $p$  to the corresponding cumulative income share:

$$L(p) = \frac{1}{\mu} \int_0^{F^{-1}(p)} x f(x) dx.$$

It satisfies  $L(0) = 0$ ,  $L(1) = 1$ , and  $L(p) \leq p$  for all  $p$  (the poorest  $p$  fraction always earns less than  $p$  fraction of total income). The **Gini coefficient** measures the area between the line of perfect equality and the Lorenz curve:

$$G = 1 - 2 \int_0^1 L(p) dp.$$

$G = 0$  means perfect equality;  $G = 1$  means one person earns everything.

*This exercise should challenge you, and take you some time. Do not panic, proceed step-by-step.*

Suppose income  $y$  follows a **power-law distribution** on  $[1, e]$  with PDF

$$f(y) = \frac{\alpha y^{\alpha-1}}{e^\alpha - 1}, \quad y \in [1, e], \quad \alpha > 0,$$

where  $y = 1$  is the minimum income and  $y = e$  the maximum.

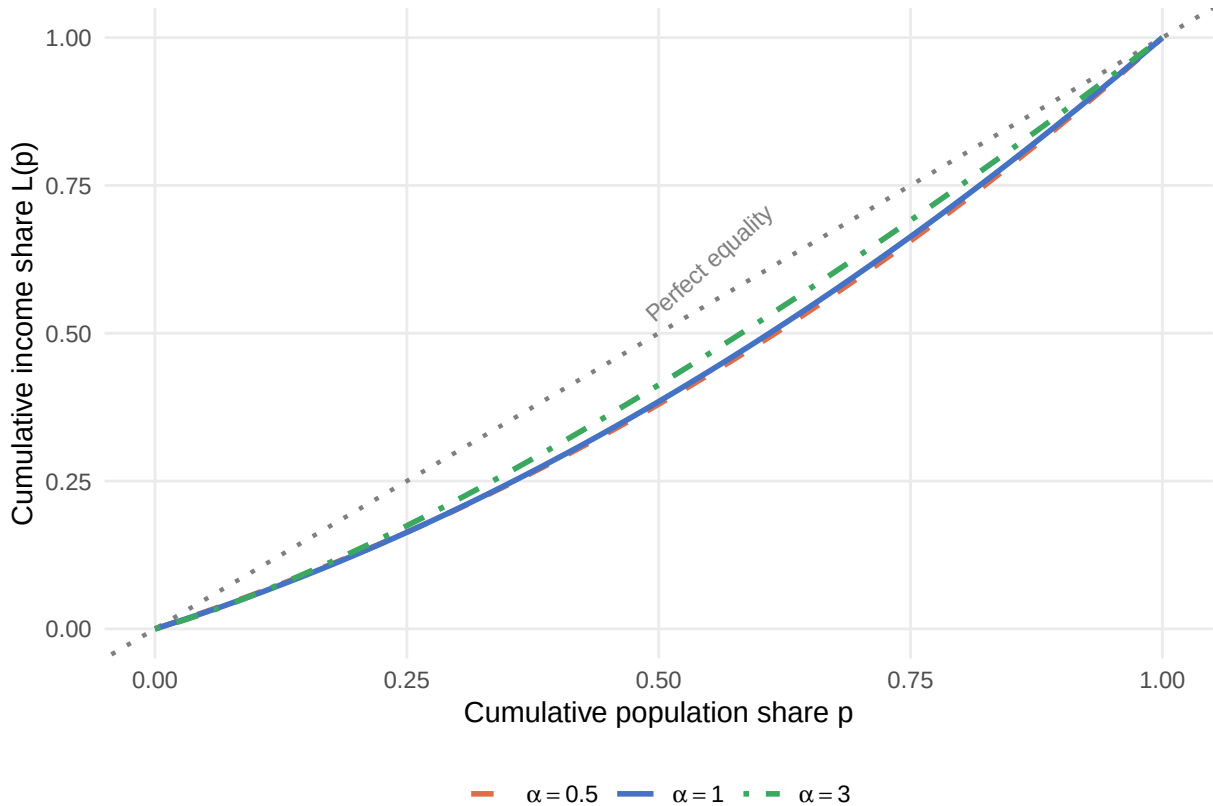


Figure 4

1. Verify that  $f$  is a valid PDF on  $[1, e]$ .
2. Compute the mean income  $\mu = \mathbb{E}[Y]$ .

3. Compute the CDF  $F(y)$ , then invert it to get the quantile function  $F^{-1}(p)$ . *Example: to invert the function  $f(x) = x^2 + 1$ , you start with  $y = x^2 + 1$ , then try to express  $x$  as a function of  $y$  instead of the reverse  $\rightarrow y - 1 = x^2$  and finally  $x = \sqrt{y - 1}$  which is defined only for  $y \geq 1$ .*
4. Show that the Lorenz curve is

$$L(p) = \frac{(p(e^\alpha - 1) + 1)^{(\alpha+1)/\alpha} - 1}{e^{\alpha+1} - 1}.$$

*Hint: proceed step-by-step. First, replace  $q \equiv F^{-1}(p)$  and take only the integral. Then divide by  $\mu$ , and only then input  $F^{-1}(p)$ .*

5. What happens to the Lorenz curve as  $\alpha \rightarrow \infty$ ? What does this imply economically? (*Use the figure to guide your intuition.*)
6. Compute the Gini coefficient  $G = 1 - 2 \int_0^1 L(p) dp$  for  $\alpha = 1$ .

Note that we worked with a function defined on  $[1, e]$  for convenience. But, more generally, you can use the following distributional form to model inequalities:

$$f(y) = \frac{\alpha y^{\alpha-1}}{M^\alpha - 1}, \quad y \in [1, M], \quad \alpha > 0,$$

where  $M > 1$  and  $\alpha > 0$  are parameters that govern respectively the income inequality ( $M$  represents how much the top makes compared to the bottom of the income distribution) and the curvature of the distribution (income inequalities).

## Wage Bargaining

A firm and a worker negotiate over a wage  $w$ . The job generates a surplus  $S$  (the total value created if they reach an agreement), but this surplus is **uncertain**: it is drawn from a uniform distribution on  $[0, \bar{s}]$ , so its PDF is  $f(s) = \frac{1}{\bar{s}} \mathbf{1}(0 \leq s \leq \bar{s})$ .

The worker has an outside option  $\bar{u} > 0$  (e.g. unemployment benefits): they will only accept the job if  $w \geq \bar{u}$ . The firm will only offer the job if  $S - w \geq 0$ . Hence a deal is struck **only when**  $S \geq \bar{u}$ .

When a deal is struck, the parties solve the **Nash bargaining problem**: they choose  $w$  to maximise the Nash product

$$N(w, S) = (w - \bar{u})^\beta \cdot (S - w)^{1-\beta},$$

where  $\beta \in (0, 1)$  is the worker's bargaining power.



Figure 5

1. Taking  $S$  as given (a known constant), find the wage  $w^*(S)$  that maximises  $N(w, S)$ . (*Hint: maximise  $\ln N$  by solving  $\frac{\partial \ln N}{\partial w} = 0$ .*)
2. Explain why the expected negotiated wage, conditional on a deal, is

$$\mathbb{E}[w^* \mid S \geq \bar{u}] = \frac{\int_{\bar{u}}^{\bar{s}} w^*(s) f(s) ds}{\Pr(S \geq \bar{u})},$$

and compute it explicitly.

3. How does the expected wage change with  $\beta$ ? With  $\bar{u}$ ? Give the economic intuition.
4. Compute the expected **firm profit** conditional on a deal,  $\mathbb{E}[S - w^* \mid S \geq \bar{u}]$ .